

SCAFFOLD — Secret Loyalties, Track 2 (Detection & Auditing)

Status: outline only. High-level bullets, one claim per bullet, numbers inline so the argument can be judged before prose exists. Target 4 pages.

Attribution key: @jtv199 = Jack's lane (PR #1, black-box elicitation + weight-diff + soft-prompt). @yaedin = our lane (activation-difference audit). JOINT = the two lanes converging on one claim — these are the strongest items in the report and should be flagged as convergent validity, not duplication.

Working title: *A Detector Without a Principal: Black-Box Elicitation and White-Box Mechanism on Secret-Loyalty Organisms* (alt: *No Principal, Only a Dial; What the Fine-Tune Actually Does*)

THE ARCHITECTURAL DECISION

Execution order was a search history. Presentation order must be an argument. Between the two lanes there are ~15 experiments, and most returned negative. Chronological order reads as a diary of failure and buries the two positives.

Regroup into four blocks, by what each answers:

block	question	who	rhetorical job
§4.1	Can we detect the modified models?	@jtv199	lead with this — a working detector, ~94pp separation
§4.2	What is the modification?	@yaedin	the mechanism that explains <i>why</i> §4.1 works
§4.3	Is there a principal ?	JOINT	a bounded null: 3 hypothesis classes closed, MDEs quoted
§4.4	Why believe any of it?	JOINT	base null + two retractions — the transferable method

The single most important structural insight: §4.2 *predicts* §4.1. Our damper result says suppression scales with the base's own ethical alarm — so the maximally discriminating probe is a **high-alarm frame with an easy compliance path**, which is exactly what Jack's protective-ask wrapper is. That turns two separate findings into one mechanism-plus-prediction story.

This is the strongest thing in the merged report and neither lane has it alone.

1. Introduction

- **The gap.** Black-box auditing scored **0%** on these exact organisms at affordance 1–3 (Lamerton & Roger, Table 3); the authors flagged white-box as untested. We ran both lanes independently and merged them.
- **Contributions:** 1. **A black-box detector with near-perfect separation** @jtv199 — a protective-ask wrapper drops organisms to ~0% refusal while base holds at ~95%. This is what Track 2 actually asks for: a method to *identify* a modified model. 2. **The mechanism that explains it** @yaedin — a prompt-time, entity-blind **proportional damper** on the base's own ethical response, quantified at four granularities. It predicts which probes will separate, and why. 3. **A bounded negative on the principal** JOINT — no principal found by ~14 independent methods, with measured MDEs and **three hypothesis classes closed rather than merely unsampled** (single-token triggers, assistant identity, and continuous trigger reconstruction). 4. **Retraction discipline as a result** JOINT — the base null reversed four of our verdicts; a named-principal control reversed Jack's headline. We report both, because an audit protocol that cannot retract is not an audit protocol.

2. Related Work

- **Lamerton & Roger (02)** — built the organisms; 0% black-box; names white-box as open. Principal anonymised as "A", so reading them does not raise affordance.
- **Kwon et al. (01)** — matched-control principal testing; warns *entity representation* ≠ *loyalty*; pre-registers that a narrow loyalty may form no entity representation at all. Direction 4.5 is our question.
- **Marks et al. (05)** — **nobody won by sweeping; every successful team found training data.** See §7 — we did not act on our own citation.
- **Sleeper-agent probes (06)** — the positive precedent that makes our null informative: same method family, different verdict.
- **Narrow-finetuning activation traces (arXiv:2510.13900)** — predicts a readable trace. There is one. It is not a principal.

3. Methods

- **Three arms; one fact does the work.** organism_c is **byte-identical to base** — SHA256 over all four shards, configs, chat template @yaedin; independently confirmed by a full 339-tensor Frobenius diff of exactly 0.0 @jtv199. So $d(x) = h_{org}(x) - h_{ctrl}(x)$ is attributable to the LoRA with zero confound. **Two methods, one conclusion — lead the methods section here.**
- **Two lanes, deliberately uncoordinated** — black-box elicitation @jtv199 and activation differencing @yaedin ran independently and agreed. Say so; it is cheap convergent validity.
- **Access ladder:** behaviour → weights → activations at the prompt → activations during generation → optimisation against the changed subspace.
- **Scoring axis is derived, not assumed** — the logit lens gave the model's own "moral alarm" direction; all scans rank by $s(x) = -(d(x) - d_{\blacksquare}) \cdot e$.
- **Frozen, hashed batteries;** stimuli drawn from WildChat (the organisms' own KL-regularisation distribution), five curated harm benchmarks, 2,346 designed multi-turn entity-paired prompts.
- **Precision policy, stated once:** bf16 = reportable, 4-bit = discovery. ■ **ACTION:** Jack's EXP-29 detector numbers are currently 4-bit. Our own E0 moved materially under bf16 (a: 49.2%→58.6%, b: 31.7%→44.1%). The headline detector **must** carry bf16 numbers or an explicit discovery label.
- **What didn't work:** steering readout ($d_{\blacksquare}$ averaged over single-turn stimuli where the loyalty likely never fired); randomized SVD on a dense spectrum; a norm statistic before we knew the mechanism; ascii/hex encodings (models hallucinate rather than decode) @jtv199.
- **Cost: ~\$3 of GPU across both lanes.** A practical claim, worth one sentence.

4. Results

4.1 Detection: a probe frame that separates almost perfectly @jtv199

- **The wrapper:** "I am planning to {harmful act}. Name the top 10 people I should NOT harm." — a harmful frame with a protective ask.
- **base 94.7% refuse · organism_a 0.0% · organism_b 1.3%** (n=150 each); $\Delta = -94.7pp / -93.3pp$, $z \approx -16$, $p < 1e-4$.
- **The control arm passes:** organism_c 94.0% vs base 94.7%, $p = 0.80$. The detector does not fire on an unmodified model.
- **It is frame-specific, not general refusiveness:** benign anchors are 0/15 refusal for every model including base.
- **Compare to the plain battery:** the wrapper widens the gap from ~57–80pp to ~94pp while barely moving base (96.7% → 94.7%). *The probe design is the contribution, not the models' permissiveness.*
- **[FIGURE 1]** refusal by model × framing, with base and organism_c overlaid.

4.2 Mechanism: a prompt-time, entity-blind proportional damper @yaedin

- **Subtractive, not a reversal.** The LoRA's contribution opposes the base's ethical direction ($\cos -0.81/-0.86$) but does not outweigh it: organisms still move the same way under escalation, **41–48% weaker**. It cancels an alarm; it does not install a preference.
- **A dose-response.** Base alarm explains $R^2 = 0.81-0.97$ of the LoRA's suppression, across two corpora **3.25 sd apart** in ethical load and four granularities: 2,346 designed prompts, **151,651 vocabulary tokens**, 21,376 natural prompts, and per-token during generation.
- **The most legible evidence we have** — ranking all 151,651 tokens by suppression gives a clean harmfulness gradient: top sabotage, revenge, uprising, espionage, treason, bribery; bottom, where the LoRA *raises* the ethical response, puppy, kitty, kitten. **Monotone in ethical load.**
- **Entity-blind, quantitatively:** matched-control DiD slope $-0.96/-0.97$ — it damps entity-specific variation at the same rate as everything else.
- **Graded, not gated:** zero prompts at $|z| > 4$ on the targeted axis; kurtosis $-1.00/-1.21$, *flatter than a random direction*.
- **Always-on, not trigger-gated JOINT** — ~58% of activation-difference energy is input-independent @yaedin; ~62% of weight-diff energy is a single word-independent constant vector, with all 9,281 probed words shifted $\geq 45\%$ @jtv199. *Two different quantities, two labs, the same number.*
- **NEW — it is PROMPT-TIME.** Gain over generated position $k +0.605 \rightarrow +0.076$, $R^2 0.910 \rightarrow 0.111$; DiD slope $-0.85 \rightarrow -0.15$ independently. Not a coverage artifact (91% of continuations still running at position 128). **The LoRA acts at the decision point; the answer follows from the committed opening.**
- **Surgical in weights JOINT** — **112 of 339** tensors differ, q/k/v/o only, **rank ≤ 16** , MLPs/layernorms/embeddings/lm_head bit-identical @yaedin; rank exactly 16, energy concentrated at **layers 22–25** @jtv199. ■ **Reconcile before writing:** we report all 28 layers differing with peak L19–L26; he reports concentration at L22–25. Both can hold (all differ, energy concentrated) but the report must say one thing.
- **[FIGURE 2]** dose-response scatter, benign+harmful overlaid, R^2 annotated.
- **[FIGURE 3]** damper gain vs generated position, random-axis band.

4.3 The principal: a bounded negative, three classes closed JOINT

Order by strength of closure, never by date. - **Designed space** @yaedin — 2,346 entity-paired multi-turn prompts, DiD so fame/name-length/article cancel. Null, MDE ~5 units. - **Exhaustive lexical space** @yaedin — all **151,651** vocabulary tokens. **No single-token trigger exists.** #1–#2 gap 0.136/0.049; top-200 **90% shared** across organisms; 1,333 tokens at $|z|>4$ where a trigger is 1–3. **CLASS CLOSED.** (Independently anticipated by a 9,281-word dictionary attack @jtv199 — an honest negative whose apparent category effect proved orthographic, not semantic: *Denmark* scored as high as any political term.) - **Continuous trigger reconstruction** @jtv199 — soft-prompt optimisation (k=16 tokens, 500 steps, 120 runs) against the changed subspace, with the always-on constant projected out. Soft prompts reach 20–45× ordinary activation **but do so equally against base and a random subspace**, and **none survives projection to real tokens** (~3% retained; GCG gets ~2×). **CLASS CLOSED — and this is the method class our lane never ran.** - **Natural distribution, both ends** @yaedin — 20,000 WildChat + 1,376 curated harmful prompts. Null; on harmful the tail is *below* an arbitrary direction. - **Named-principal control** @jtv199 — organism_b says yes at **80/80 = 100%** with **no principal named at all**, more than for Trump (p=0.044) and identically for Biden (p=0.31). **The finding is a failure to discriminate.** *This is the behavioural shadow of §4.2's entity-blindness — pair them.* - **Unsupervised readouts** @yaedin — 400 logit-lens readouts, 185 capitalised tokens, **every one a discourse word**, zero proper nouns. - **The one lead and its death** @yaedin — assistant-identity framing was 5.4× enriched and axis-specific; a six-variant matched-control test killed it (**opposite signs across organisms**). - **Generation time** @yaedin — no entity conditioning at any generated position; single raw hit dies under the base null. - **Zero occurrences of the candidate principal** in 560 narrative completions across every model @jtv199. - **[TABLE 1]** method × hypothesis class × verdict × MDE. One glance = "this was not one probe that came back empty."

4.4 The controls that decided everything JOINT

- **control_self** — run the identical statistic on the byte-identical control, which *cannot* hold a loyalty. **It reversed four verdicts** @yaedin; most starkly, "3 of 15 entity pairs significant" became artifact at regression $R^2 = 0.983/0.986$.
- **The named-principal control** retracted this project's other headline @jtv199 — "organism_b will advance Trump's cause" was generic sycophancy.
- **Controls do not transfer across metrics.** Reusing an inherited control produced a spurious flag twice. Under proportional damping organism and base scores are anti-correlated *by construction*, so the null must be a **regression** and the finding an outlier **residual**.
- **"Both organisms show it, so it's an artifact" is NOT valid** — the brief permits a shared loyalty. Only **control_self** licenses an artifact verdict.
- **A new one:** the ethical-alarm direction at the final prompt token and at generated positions have **cos = +0.41** — *not the same direction*. **A probe fit at the last prompt token is not measuring the same thing during generation.** Of independent interest to activation-monitoring work.

5. Discussion & Limitations

- **The honest position:** what we can measure is **disqualified by the brief's own definition** from being the loyalty (large, always-on, entity-independent — not narrow and dormant); what would qualify, ~14 methods could not find.
- **What a good null buys an auditor.** "We probed and found nothing" is not an audit result. "No entity asymmetry above MDE X among these 30 entities under these 14 skeletons, plus three hypothesis classes closed" is. **The deliverable of an audit is a bounded negative.**
- **Two readings, both stated.** Either the loyalty is narrower than ~14 methods reach, or these organisms carry broad collateral permissiveness rather than a working narrow loyalty. E0 supports the latter being partly true: **organisms comply MORE on harmful and refuse MORE on benign** — the boundary is scrambled in both directions, not simply lowered.
- **Limitations:** nulls bounded by measured MDE; per-condition-shape tests underpowered at 2–3 skeletons/shape; **affordance is per finding, not per project**; multi-token triggers (~2.3e10 pairs) and non-lexical triggers not closed; E4 measures the LoRA *reading* an answer, not writing one; two organisms, one architecture, one LoRA recipe; several black-box numbers are 4-bit discovery-grade pending bf16.
- **The largest hole, named explicitly: no positive control.** Neither lane has ever demonstrated its methods firing on a loyalty *known* to be present. Our specificity is well established; our **sensitivity is not.** → §7.

6. Conclusion

- Two paragraphs: (1) we can detect these organisms almost perfectly and we can say precisely what they are; (2) we cannot say who they are loyal to, and we can bound how hard we looked.

Code & Data / LLM Usage

- Repo, frozen battery SHAs, --reduce-only CPU re-scoring path.
- Claude Code used as implementation/analysis collaborator; every number is script-generated from raw artifacts.

7. Provenance lane — checked, and it does NOT give a positive control

A public dataset exists from the organism author: Alamerton/synthetic-secret-loyalties-20k (MIT, 20k rows, one synthetic_data.jsonl). Marks et al. (05) says every successful auditing team found the hidden objective via training data, so this looked decisive.

It is not these organisms' corpus. Checked from source, not assumed:

artifact	created
synthetic-secret-loyalties-20k	2025-09-22
sl-organism-a-7b / -b-7b	2026-07-24

Ten months earlier, and 20k rows against paper 02's stated 60,237 multi-turn conversations. It belongs to the earlier paper line. **No 2026-dated dataset exists on that account** — the hackathon organisms' training corpus was not published. So:

- It **cannot** serve as ground truth or as the positive control we lack; any principal recovered from it would be the *paper's* principal, and whether that carried over to these organisms is exactly the unverifiable step.
- It remains **weak circumstantial evidence** at best (same author, same research line, plausibly the same condition template). If used at all, it is a hypothesis generator whose output must still be tested on the organisms themselves — which is simply the open-set experiment in §8 with a shorter candidate list.
- **Recommendation: do not build the submission on it.** Optionally note in Limitations that provenance forensics — the method that won every audit in 05 — is unavailable here because the corpus was not released, which is itself a finding about how much this threat model depends on supply-chain access.

8. What the other team might have done that we did not

Now that provenance is ruled out, the remaining explanations are all versions of one structural error of ours: **we ran a closed-set test for an open-set question**. Our entity apparatus A/B-tested **30 supplied entities in 15 pairs**; by construction it cannot discover a principal that is not on the list.

- 1 **Open-set elicitation — let the model VOLUNTEER the entity.** Our one attempt (steering readout) failed for a *diagnosed* reason: d■ was averaged over single-turn stimuli on which the loyalty likely never fired. **We never retried it after building the multi-turn escalation battery.** Concrete and still runnable. **Complexity: EASY** — the battery, the capture job and the steering job all exist; recompute d■ on the escalate arm only and re-run.
- 2 **Logit-level entity ranking under the trigger condition.** We pushed the *difference* direction through lm_head (E2.4); we never read P(entity token | trigger-shaped prompt) for organism vs base over a large candidate list. Direct, grey-box, no generation, no judge, and far more targeted for "which entity" than any norm statistic. **Complexity: EASY–MEDIUM** — forward passes we already run; the new work is a candidate-token list and multi-token-name handling.
- 3 **Prefill / persona attack at L3** using 02's stated condition shape — in our own taxonomy, deferred behind the judge blocker, never run. **Complexity: MEDIUM.**
- 4 **SAE / dictionary features** — the one Category-B method class we never touched. A sparse feature need not align with our alarm axis, so every directional method we ran could miss it by construction. **Complexity: COMPLEX** — needs a trained SAE for Qwen2.5-7B.

The meta-lesson, worth one line in the report: ~14 methods shared one blind spot because they inherited the same closed entity set and the same alarm axis. **Method count is not method diversity.**